

DAVI Antony

The Braess paradox, build more for more mobility?

The analysis of the impact of human behavior on road flows allows a better understanding of mobility issues. At a time when the facilitation of transport favors directly economic, social and ecological competitiveness, the analogy of the model studied towards other types of flows (air, financial, etc.) can also be used.

This project is the subject of group work.

List of band members:

- GAUTIER DE LA PLAINE Guillaume
- ROUSSEAU Roman
- DAVI Antony

Thematic positioning (phase 2)

COMPUTER SCIENCE (Practical Computing).

MATH

Key - Words

FRENCH	ENGLISH
FLUX	<i>Flow</i>
OPTIMISATION	<i>Optimization</i>
MODÈLE	<i>Model</i>
GRAPHE	<i>Graph</i>
RÉSOLUTION	<i>Resolve</i>

Annotated bibliography :

Traffic problems in the city, or more generally on a road network, are a recurring phenomenon. There are several results concerning them, in particular the Paradox of Braess, highlighted by the German mathematician Dietrich Braess in 1968. It was observed several times since, and for one of the first times "In 1990, on the occasion of the day of the Land, [when] the City of New York decided to close 42nd Street to traffic," as This is explained in the CNRS article [1], at the origin of our subject. Since New York's 42nd Street being one of the busiest, logic would have dictated that traffic was generally worse in the sector, regular users of 42nd Street having to use another route. However, as this article explains, the opposite happened, the circulation was made more fluid. It's a paradox which is only possible on the assumption that each user chooses his path in a selfish , that is to say without taking other users into account.

It was not the design of the skeleton of the algorithm that required real documentation but the garrison that counts the graphic display, the initial conditions, or even the problems

convergence of the model. A study of a problem on graphs cannot go without a minimum theoretical course. This is why we have therefore procured two courses on the theory graphs.

One by Michel Rigo [2], professor at the University of Liège, level L2, the other by Hervé Barbot [3], Doctor in Computer Science. Their content allowed us to have a less naive approach to the issue, in particular on the convergence of the device towards a stable state (how to avoid an overload of all ? What order of magnitude would our initial conditions be? ...) as well as on the realization of the paradox. We will opt for grids of 10 to 20 cities.

The use of this book [4] allowed the realization of our ambitions on the manipulation of large amount of data inherent in the initial conditions, in particular on the use of dictionaries facilitating access to information during a given process. It was particularly useful concerning Dijkstra's algorithm and its complexity which must be kept to a minimum in order to study bulky systems therefore more relevant.

Finally, beyond graphic modeling, manipulation of the algorithm requires the use of adapted libraries, it is Tkinter which was retained in order to be able to manage the iterations and to detect program convergence flaws. The language being complex, we used the associated documentation [5].

Selected issue:

As we have seen, the Braess Paradox can be a way to solve the problems of circulation, this under certain hypotheses and under certain conditions.

So, how can Braess's paradox be used to solve traffic problems?

Objectives of the project:

- Computer modelling of the evolution of automobile flows within a road network
- Follow-up of the evolution and modification of the system in order to obtain a "stable" state in which the attributes associated with the data vary little over time, then evaluation of the "time of average route".
- Then recursive study of the slightly modified system (by amputation of a random route) in the hope of obtaining an average travel time lower than that of the system previous.
- Comparison of social and selfish routing.
- Analysis of several traffics theorem (Price of anarchy ...)

Distribution of tasks :

- *My work is mainly in the graphic modelling and the development of the core of the model. I remastered the project by making readable and class oriented in 2022.*
- *Romane worked on useful functions required to generate valuable graphs and coherent data.*
- *Guillaume worked on the tkinter interface.*

Bibliographic references :

- [1] Étienne Ghys : Price of anarchy: <https://images.math.cnrs.fr/Le-prix-de-l-anarchie.html>
- [2] Michel Rigo : Graph Theory : http://www.discmath.ulg.ac.be/cours/main_graphes.pdf
- [3] Hervé Barbot: Graph Theory: <http://www.momirandum.com/graphes/theorie-desgraphes/TheoriedesGraphes.html>
- [4] Lionel Uhl : 1001 codes python (for CPGE),
- [5] Fredrik Lundh: An Introduction To Tkinter: <http://effbot.org/tkinterbook/tkinter-hellotkinter.html>

GitHub Link : https://github.com/kOraty/Braess_Paradox